# CS 215 Midsem, Fall 2020

**INSTRUCTIONS:**

1. Attempt all six questions. Clearly mark out rough work, mention question numbers. Write your name and roll number on the top of the first page.

2. You are allowed to refer to all class notes, slides, handouts and the class textbook as well as previous years quizzes/midsems that I shared with you. You are not allowed to browse the web or talk to each other. **By submitting this midsem, you promise that you abided by these policies.**

3. You may use any results or theorems discussed in class directly, without proving/deriving them from scratch.

4. When in doubt, make suitable assumptions and state them clearly. Do not ask proctors any doubts pertaining to the questions.

5. ~~Show your ID card to the proctor when asked (between 5:30 and 6:00 pm).~~

6. Record your screen using the chrome screen recorder software in lowest possible resolution (240 p).

7. Write all answers neatly with black or blue ink. Take clear pictures of all your work in the correct order and create a zip file or a PDF (preferable, as far as possible). Name the file as YourRollNumber.zip/.pdf.

8. Your submission on moodle needs to be uploaded before 9:00 pm. The submission consists of (1) the actual exam PDF/zip, as well as (2) a text file (named as YourRollNumber.txt) containing a link to the google drive folder where you will upload your screen video. Do not upload the screen video on moodle, but only on the google drive folder.

9. You have 2.5 hours for the midsem (up to 8:30 pm). You should **stop writing** at 8:30 pm. You have up to 30 minutes more for scanning and uploading your work on moodle. No submissions will be accepted after that.

10. You have to upload your screen video by 6:00 pm on 8th October on your own google drive folder (the link to which you included in the text file).

**QUESTIONS:** *Please read the instructions above first.*

1. Let $X_1, X_2, ..., X_n$ be iid random variables from $\mathcal{N}(\mu, \mu)$, i.e. a Gaussian distribution with mean $\mu$ and variance $\mu$. Determine $\mu$ by maximum likelihood. [15 points]

2. Consider a set of independent random variables $X_1, X_2, ..., X_n$ from a distribution with parameter $\alpha$. We say that an estimator $V$ is said to be **apposite** for the parameter $\alpha$ if the conditional distribution of $X_1, X_2, ..., X_n$ given any value $V = v$ is independent of $\alpha$ (i.e. if the formula for the conditional distribution does not contain any terms in $\alpha$). Now suppose $X_1, X_2, ..., X_n$ are the respective results of $n$ independent tosses of a coin with heads probability $p$, i.e. for any $i \in \{1, 2, ..., n\}$, we have $X_i = 1$ if you get a heads (which happens with probability $p$) and 0 otherwise. Let $\tilde{V} = \sum_{i=1}^n X_i$. Determine with appropriate justification using the conditional distribution whether $\tilde{V}$ is apposite for parameter $p$. [15 points]

3. Let $X$ be a random variable in such a way that $\log X \sim \mathcal{N}(\mu, \sigma^2)$. Find the PDF, CDF, mean, mode and median of $X$. [15 points]

4. Consider $n$ independent indentically distributed random variables $\{Z_i\}_{i=1}^n$ such that each $Z_i$ takes on the value $-1$ or $+1$ with 50% probability. Let $c_1, c_2, ..., c_n$ be real numbers. Here, we wish to prove that for any $t > 0$, we have $P(\sum_{i=1}^n c_i Z_i \geq t) \leq \exp\Big(-t^2/(2\sum_{i=1}^n c_i^2)\Big)$. To this end, do the following:

   (a) Prove that $P(\sum_{i=1}^n c_i Z_i \geq t) \leq e^{-\lambda t} E\Big[\exp\Big(\lambda \sum_{i=1}^n c_i Z_i\Big)\Big]$, where $\lambda > 0$ is a fixed parameter with which both sides of the inequality $\sum_{i=1}^n c_i Z_i \geq t$ are multiplied. We will optimize on $\lambda$ later.

(b) Show that the RHS of the result in part (a) is equal to $e^{-\lambda t}\prod_{i=1}^{n}\cosh(\lambda c_i)$, where $\cosh(x) = (e^{-x}+e^{x})/2$ is the hyperbolic cosine, and where $\prod$ denotes a product.

(c) Using the fact that $\cosh(x) \le e^{x^2/2}$ for any real-valued $x$, show that $P(\sum_{i=1}^{n} c_i Z_i \ge t) \le \exp\Big(-\lambda t + \lambda^2 \sum_{i=1}^{n} c_i^2/2\Big)$.

(d) Optimize the bound in part (c) w.r.t. $\lambda$ and complete the proof. [4+4+2+5=15 points]

5. Consider that $X_1, X_2, ..., X_n$ are $n$ independent and identically distributed random variables, each belonging to the density $f_X(x;\theta)$ with parameter $\theta$. Let $\hat{\theta}$ be the maximum likelihood estimate of $\theta$. For a one-one function $g$, prove that $g(\hat{\theta})$ is the maximum likelihood estimate of $g(\theta)$. What happens if $g$ is many-one? [12+3=15 points]

6. Let $Y$ be a continuous random variable that is uniformly distributed from $-1$ to $+1$. Consider a sequence $Y_1, Y_2, ...$ of independent identically distributed random variables with the same distribution as $Y$. Consider the following:
$U_i = \sum_{j=1}^{i} Y_j/i$ (i.e. the average of the first $i$ random variables in the sequence),
$W_i = \max(Y_1, Y_2, ..., Y_i)$ (i.e. the maximum value of the first $i$ random variables in the sequence),
$V_i = \min(|Y_1|, |Y_2|, ..., |Y_i|)$ (i.e. the minimum absolute value of the first $i$ random variables in the sequence),
$H_i = \prod_{j=1}^{i} Y_j$ (i.e. the product of the first $i$ random variables in the sequence).

For any small-valued $\epsilon > 0$ (small-valued means $\epsilon \approx 0$), determine the following:
$\lim_{i\to\infty} P(|U_i - E(U_i)| > \epsilon)$,
$\lim_{i\to\infty} P(|W_i - 1| > \epsilon)$,
$\lim_{i\to\infty} P(|V_i| > \epsilon)$.
From the last result, determine $\lim_{i\to\infty} H_i$.[15 points]